

Daily Science Anecdotes

Topic: Your Brain Is Lying to You!

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September 17, 2026

Right now, you are not seeing the world. You are seeing a **movie your brain is making up**, and it is very good at hiding that from you.

Try this: hold your arm out and look at your thumbnail. That tiny patch, only about 2° of your vision, is the *only* place where you see sharply. Everything else is a blurry, almost colorless mess. So why does the whole room look crisp and colorful? Because your eyes dart around and your brain **paints in the rest**. Wooooow, right?

Now for the creepy part. Each of your eyes has a **hole** in it. Where the optic nerve leaves the retina, there are no light-sensing cells at all. The hole is about 5° to 6° wide, roughly *ten full moons side by side*, and yet you have never seen a black spot. Your brain looks at the surroundings, guesses what should be there, and fills it in, the same way you would estimate a missing point on a graph. It deletes a piece of reality and hides the edit!

And it does not stop there. Your eyes jump about **3 times every second** (these jumps are called saccades), and during each one your brain switches vision off so the world does not blur. If each blackout lasts about 0.04 seconds, that is $3 \times 0.04 = 0.12$ seconds of darkness in every single second. Over a day, that adds up to **an hour or more of blindness** that you never noticed. Not once. Prove it in five seconds! Stand in front of a mirror, look at your left eye, then shift your gaze to your right eye. You will never see your eyes move: you see one eye, then the other, with nothing in between. Now film yourself doing the same thing and play the video back. **Your eyes clearly jump!** Your brain edited the movement out of your life.

Even your fear is math. You walk into a dark room and see a person in the corner. Your heart pounds. A second later, it is just a jacket. Suppose the chance of an intruder is only 1%, and a human-like shape shows up 90% of the time for a person but 30% of the time for a jacket. Bayes' theorem says the shape is a person only about **3%** of the time, so it was almost certainly a jacket. But your brain panicked anyway, because it is not built to be right, it is built to **survive**. It works like a smoke detector that would rather scream at toast than miss a fire.

Even a simple drawing can break you. A Necker cube, just a flat wireframe, flips between two 3D shapes, and you cannot stop it. Turning a flat 2D image back into a 3D world has infinitely many possible answers, so your brain keeps picking one, then the other. Scientists call this an *ill-posed problem*. You call it “wait, what?!”

The leading theory, called **predictive processing**, says your brain does not record reality. It **predicts** it, then corrects itself whenever the senses disagree. Neuroscientists describe what you experience as a **controlled hallucination**, kept in check by the real world. (The “Bayesian brain” is still a hypothesis, not proven fact: the brain behaves *as if* it does this math.)

So the world is real. But what you see is your brain’s best guess about it.

**So, if your brain can hide a hole in your vision,
what else has it been hiding from you?**